

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO2 ASSESSMENT TOOL 1 – Scenario-Based Requirement Analysis

Course Code:	CSA10 / CSA1063	Course Title:	Software Engineering
CO Assessed:	CO2	Assessment:	Assessment Tool 2 – Scenario-Based Requirement Analysis
Weightage:	20%	Total Marks:	25
CO2 Statement:	<i>Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills</i>		
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Date:	03-08-2026	Duration:	60 minutes

Code of Conduct

I _____ jeyakannan L _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Annexure A – Scenario-Based Requirement Analysis

Question

Based on the **assigned problem statement**, perform a **Scenario-Based Requirement Analysis** and prepare a Software Requirement Specification (SRS) document. Your analysis should include the following:

1. Study and Analyse the Scenario:

1.1 Carefully understand the given problem statement.

Problem Statement

The **Fraud Detection and Financial Transaction Monitoring System** is a banking and security application designed to identify fraudulent financial transactions in real time. Traditional banking systems often rely on manual verification or basic rule-based checks, which may fail to detect sophisticated fraud patterns. This project aims to develop an intelligent monitoring system that continuously analyzes transaction activities, detects suspicious behavior, minimizes financial losses, and enhances the security of digital banking services.

1. STUDY AND ANALYZE THE SCENARIO

Fraud Detection and Financial Transaction Monitoring System



1.2 Identify the business domain, stakeholders, existing challenges, and system objectives.

DOMAIN:

Aspect	Details
Domain	Banking and Financial Security
Industry	Banking, FinTech, Digital Payment Systems
Application Type	Fraud Detection & Transaction Monitoring
Primary Purpose	Detect fraudulent transactions and protect customer accounts
End Users	Banks, Financial Institutions, Customers, Security Teams

Stakeholders:

Stakeholder	Role in the System
Bank Customers	Perform financial transactions securely.
Bank Administrators	Monitor system activities and manage user accounts.
Fraud Investigation Team	Reviews suspicious transactions and takes appropriate action.
Security Analysts	Analyze fraud patterns and improve detection rules.
Bank Management	Monitors fraud reports and ensures regulatory compliance.
Regulatory Authorities	Ensure banking operations comply with financial security regulations.

Existing Challenges

- Manual verification is time-consuming.
- Fraudulent transactions are sometimes detected only after financial loss.
- High transaction volumes make continuous monitoring difficult.
- Traditional rule-based systems generate false alerts.
- New fraud techniques continuously evolve, making detection challenging.
- Delayed response increases customer dissatisfaction and financial risk.
- Lack of real-time monitoring reduces banking security.

System Objectives

Objective	Description
Real-Time Monitoring	Continuously monitor every financial transaction.
Fraud Detection	Identify suspicious and abnormal transaction patterns.
Risk Assessment	Assign risk levels based on transaction behavior.
Alert Generation	Notify administrators immediately when suspicious activity is detected.
Reduce Financial Loss	Prevent unauthorized transactions before completion.
Improve Banking Security	Protect customer accounts and sensitive financial information.
Support Decision Making	Provide reports and analytics for fraud investigation teams.

Scenario Analysis Summary

Component	Analysis
Business Domain	Banking and Financial Security
Main Problem	Increasing fraudulent financial transactions and delayed fraud detection
Target Users	Customers, Bank Administrators, Security Analysts, Fraud Investigation Teams
Major Challenges	Manual monitoring, false positives, delayed detection, evolving fraud techniques
Expected Solution	An intelligent fraud detection and transaction monitoring system capable of identifying suspicious transactions in real time
Expected Benefits	Improved security, reduced financial losses, faster fraud response, enhanced customer trust

2. Identify Functional and Non-Functional Requirements

2.1 List all functional requirements required by the proposed system.

Requirement ID	Functional Requirement	Description
FR1	User Registration and Login	Allow customers and administrators to securely log in using authenticated credentials.
FR2	User Profile Management	Enable users to update personal information and account details.
FR3	Transaction Processing	Record and process financial transactions such as deposits, withdrawals, transfers, and online payments.
FR4	Real-Time Transaction Monitoring	Continuously monitor all financial transactions as they occur.
FR5	Fraud Detection	Analyze transactions to identify suspicious or fraudulent activities based on predefined rules or AI models.
FR6	Risk Score Generation	Assign a fraud risk score to every transaction.
FR7	Suspicious Transaction Alerts	Automatically notify administrators when suspicious transactions are detected.
FR8	Transaction Verification	Allow administrators to approve, reject, or temporarily hold suspicious transactions.
FR9	Fraud Investigation	Maintain complete transaction history for fraud analysis and auditing.
FR10	Dashboard and Reports	Display fraud statistics, transaction summaries, and security reports.
FR11	Search and Filter	Search transactions based on customer ID, account number, date, amount, or status.
FR12	Audit Log	Record all user activities and system actions for accountability and compliance.

FR13	Notification System	Send SMS, email, or in-app notifications for suspicious activities and account updates.
FR14	Role-Based Access Control	Provide different access permissions for customers, administrators, and security analysts.
FR15	Data Backup and Recovery	Automatically back up transaction records to prevent data loss.

2.2 Identify suitable non-functional requirements such as performance, security, reliability, usability, scalability, maintainability, and availability

- **Performance:** Real-time transaction processing with low response time.
- **Security:** Encryption, authentication, authorization, and audit logging.
- **Reliability:** Accurate fraud detection with fault tolerance and data integrity.
- **Usability:** Simple, responsive, and user-friendly interface.
- **Scalability:** Support for increasing users and transaction volumes.
- **Maintainability:** Modular design, reusable code, and easy maintenance.
- **Availability:** 24×7 system uptime with backup and disaster recovery.

3. Identify Actors and Use Cases

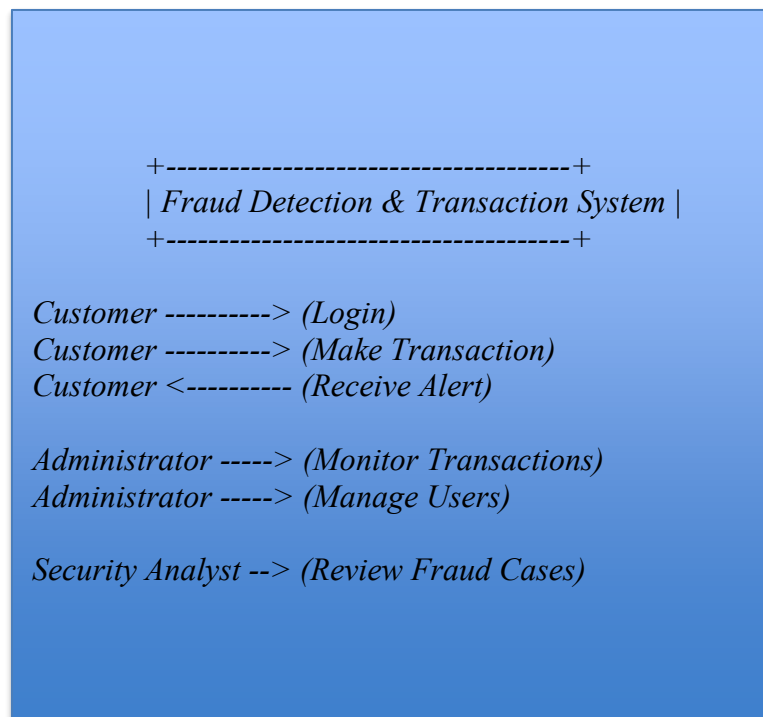
3.1 Identify all primary and secondary actors interacting with the system.

Actor	Type	Role
Customer	Primary	Performs banking transactions and receives fraud alerts.
Bank Administrator	Primary	Monitors transactions, manages users, and reviews fraud cases.
Security Analyst	Primary	Investigates suspicious transactions and updates fraud detection rules.
Notification Service (Email/SMS)	Secondary	Sends alerts and notifications to users.
Payment Gateway/Bank Database	Secondary	Processes and stores transaction data.

3.2 List the major use cases associated with each actor.

Actor	Major Use Cases	Actor
Customer	Register/Login, Perform Transactions, View Transaction History, Receive Fraud Alerts	Customer
Bank Administrator	Manage Users, Monitor Transactions, Generate Reports, Approve/Block Transactions	Bank Administrator
Security Analyst	Analyze Fraud Cases, Review Suspicious Transactions, Update Detection Rules	Security Analyst
Notification Service	Send SMS/Email Alerts	Notification Service
Payment Gateway/Bank Database	Process Transactions, Store Transaction Records	Payment Gateway/Bank Database

3.3 Draw a Use Case Diagram representing the interactions between actors and the system.



4 Prepare the Software Requirement Specification (SRS)

4.1 Introduction

The **Fraud Detection and Financial Transaction Monitoring System** is a banking security application designed to monitor financial transactions and detect fraudulent activities in real time. The system improves banking security, reduces financial losses, and protects customer accounts by identifying suspicious transactions.

4.2 Problem Description

Banks process thousands of transactions every day, making manual fraud detection difficult. Existing systems may fail to identify advanced fraud techniques, leading to financial loss and security risks. This system provides automated monitoring and fraud detection.

4.3 Scope of the System

- Monitor financial transactions in real time.
- Detect suspicious and fraudulent activities.
- Generate alerts for suspicious transactions.
- Allow administrators to review and manage fraud cases.
- Maintain transaction history and reports.

4.4 Functional Requirements

Requirement ID	Description
FR1	User Login and Authentication
FR2	Process Financial Transactions
FR3	Monitor Transactions in Real Time
FR4	Detect Fraudulent Transactions
FR5	Generate Fraud Alerts
FR6	Generate Reports

4.5 Non-Functional Requirements

Category	Requirement
Performance	Fast transaction processing
Security	Data encryption and secure authentication
Reliability	Accurate fraud detection
Usability	User-friendly interface
Scalability	Support increasing users and transactions
Availability	24×7 system availability

4.6 Use Case Descriptions

Actor	Use Cases	Actor
Customer	Login, Make Transaction, View Alerts	Customer
Administrator	Monitor Transactions, Manage Users, Generate Reports	Administrator
Security Analyst	Review Fraud Cases	Security Analyst

4.7 Data Requirements

Data Type	Description	Data Type
Customer Data	Name, Account Number, Contact Details	Customer Data
Transaction Data	Transaction ID, Amount, Date, Status	Transaction Data
Fraud Records	Risk Score, Fraud Status, Alerts	Fraud Records
User Data	Username, Password, Role	User Data

4.8 External Interface Requirements

- **User Interface:** Web-based dashboard for customers and administrators.
- **Hardware Interface:** Computer, Server, Internet connection.
- **Software Interface:** Database Management System (DBMS), Banking APIs.
- **Communication Interface:** Secure HTTPS and Email/SMS notification services.

4.9 System Features

- Secure user authentication.
- Real-time transaction monitoring.
- Fraud detection and risk analysis.
- Automatic fraud alerts.
- Transaction history management.
- Dashboard and report generation.
- Role-based access control.

5 Identify Assumptions and Constraints

5.1 List the assumptions considered while developing the system.

- Users have valid bank accounts and secure login credentials.
- Internet connectivity is available for online transactions.
- The bank database stores accurate and up-to-date customer information.
- The fraud detection rules are regularly updated.
- Authorized administrators monitor suspicious transactions.

5.2 Identify technical, operational, economic, legal, and time-related constraints affecting the project.

Constraint Type	Description
Technical	Requires stable internet, secure database, and reliable server infrastructure.
Operational	Users and administrators must follow banking policies and security procedures.
Economic	Development and maintenance costs should remain within the project budget.
Legal	The system must comply with banking regulations and data privacy laws.
Time	The project must be completed within the specified development schedule.

6 Validate Requirement Completeness and Consistency

6.1 Verify whether all stakeholder requirements have been addressed.

Validation Check	Status
Stakeholder requirements addressed	✓ Yes
Functional requirements complete	✓ Yes
Non-functional requirements complete	✓ Yes
Use cases identified	✓ Yes
Data and interface requirements included	✓ Yes

6.2 Check for ambiguity, redundancy, inconsistency, and missing requirements.

Aspect	Observation
Ambiguity	Requirements are written clearly and are easy to understand.
Redundancy	No duplicate requirements are identified.
Consistency	Functional and non-functional requirements are consistent with the system objectives.
Missing Requirements	Future enhancement can include AI-based fraud prediction and biometric authentication.

6.3 Suggest improvements to enhance the quality and completeness of the requirement specification.

- Improve fraud detection using **AI/Machine Learning** techniques.
- Add **Multi-Factor Authentication (MFA)** for better security.
- Implement **real-time notifications** via SMS and email.
- Include **detailed analytics dashboards** for administrators.

